Weekly Test — Programming

Closed Book | Time: 45 min

Arrays, Strings & 2D Array Execution

Student Name: ---------- **Date:** ---------- **Score:** ---------- / 30

SECTION A: PREDICT OUTPUT (5 SNIPPETS — 4 PTS EACH)

Analyze each code segment carefully and write the exact output produced by execution in the designated answer box.

Snippet 1: String Manipulation

```
String text = "DEVELOPMENT";
String sub = text.substring(3, 7);
char c = text.charAt(1);
System.out.println(sub + "_" + c);
```

OUTPUT:

Snippet 2: 1D Array Mutation

```
int[] nums = {2, 5, 8, 11, 14};
for (int i = 0; i < nums.length - 1; i++) {
    nums[i] = nums[i + 1] - nums[i];
}
System.out.println(nums[0] + " " + nums[2] + " " + nums[4]);
```

OUTPUT:

Snippet 3: String Array Search & Concatenation

```
String[] words = {"cat", "elephant", "dog", "giraffe"};
String result = "";
for (String w : words) {
    if (w.length() > 4) {
        result += w.charAt(0);
    }
}
System.out.println(result.toUpperCase());
```

OUTPUT:

Snippet 4: Array Element Swapping

```
int[] arr = {10, 20, 30, 40, 50};
for (int i = 0; i < arr.length / 2; i++) {
    int temp = arr[i];
    arr[i] = arr[arr.length - 1 - i];
    arr[arr.length - 1 - i] = temp;
}
System.out.println(arr[1] + ", " + arr[3]);
```

OUTPUT:

Snippet 5: String & Index Logic

```
String str = "PROGRAMMING";
int val = 0;
for (int i = 0; i < str.length(); i += 3) {
    val += i;
}
System.out.println("Val: " + val);
```

OUTPUT:

SECTION B: 2D ARRAY PAPER TRACING (10 PTS)

Trace the state changes of `matrix` step-by-step. Show variable values (`r`, `c`, `sum`) and draw the final 2D array state.

```
int[][] matrix = {
    {1, 2, 3},
    {4, 5, 6},
    {7, 8, 9}
};
int sum = 0;
for (int r = 0; r < matrix.length; r++) {
    for (int c = 0; c < matrix[r].length; c++) {
        if (r == c) {
            matrix[r][c] *= 2;
            sum += matrix[r][c];
        }
    }
}
System.out.println("Sum: " + sum);
```

Show Variable Tracing & Final Matrix State Below: